Deterministic Modelling

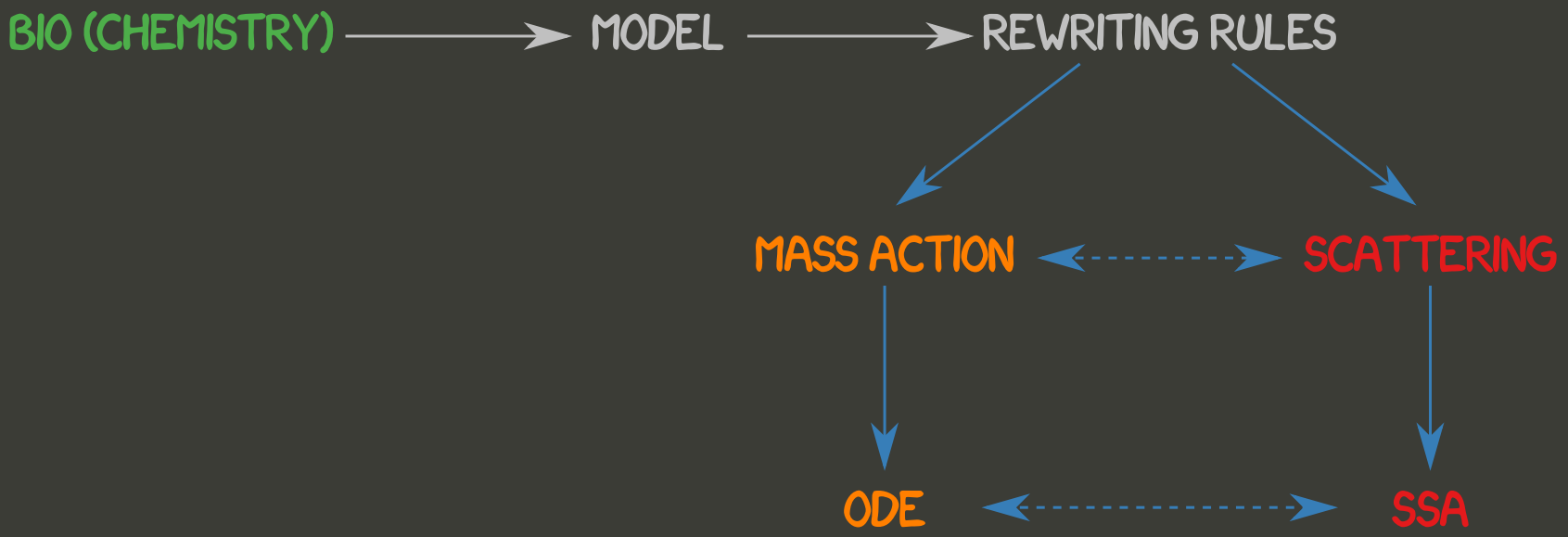
Dario Pescini[1]

Università degli Studi di Milano-Bicocca

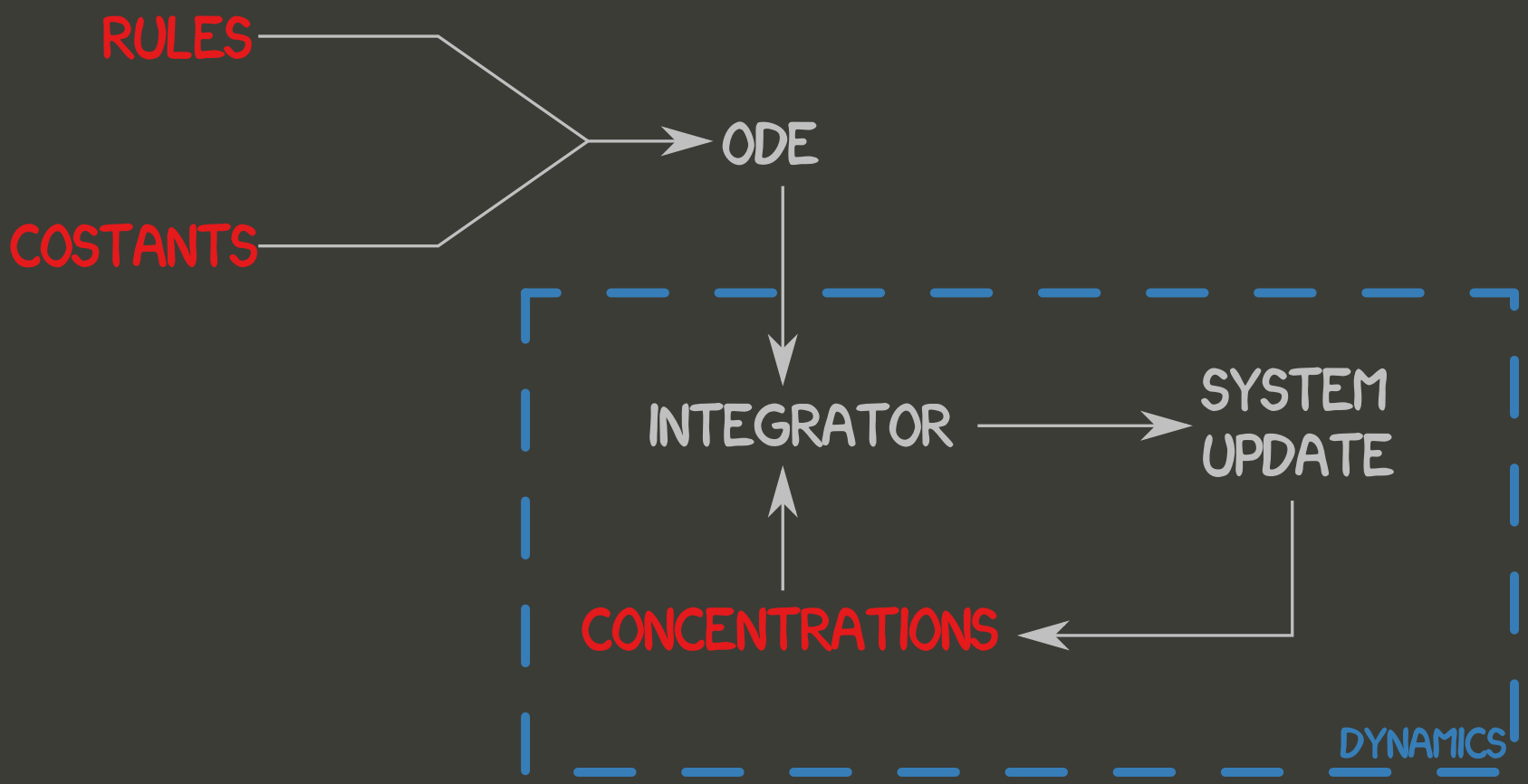
Dipartimento di Statistica e Metodi Quantitativi

dario.pescini@unimib.it

Outline


BIO (CHEMISTRY)
MODEL
REWRITING RULES
MASS ACTION
SCATTERING
ODE
SSA

Outline


RULES
COSTANTS
ODE
INTEGRATOR
SYSTEM UPDATE
CONCENTRATIONS
DYNAMICS

ODE

Systems of Ordinary Differential Equation can be used whenever it is possible to establish a functional relationship between a function and its derivative.

- Given the system variation over time, ODEs reconstruct the system behaviour
- The Cauchy problem:

$$\begin{cases} \frac{dy(t)}{dt} = f(y(t), t) \\ \\ y(t = 0) = y_0 \end{cases} \Longrightarrow \qquad y(t) = ?$$

ODE: integrator

There exist many numerical approaches to solve ODEs, and the most simple is the Euler's method:

$$\frac{dy(t)}{dt} = f(y(t), t) \Longrightarrow dy(t) = f(y(t), t)dt$$

by means of direct integration in t_0, $t_0 + dt$ we have:

$$\begin{aligned}
y(t_1) &= y_1 = y(t=0) + f(y(t=0), t)dt \\
y(t_2) &= y_2 = y_1 + f(y_1, t)dt \\
y(t_3) &= y_3 = y_2 + f(y_2, t)dt \\
&\vdots \\
y(t_{n+1}) &= y_{n+1} = y_n + f(y_n, t)dt
\end{aligned}$$

Euler's method: example

Given:

$$\begin{cases} \frac{dy(t)}{dt} = -k\,y(t) \\ y(t=0) = 100 \end{cases}$$

with analytical solution $y(t) = 100\,e^{-k\,t}$

Parameters:

$\Delta t = 0.1$

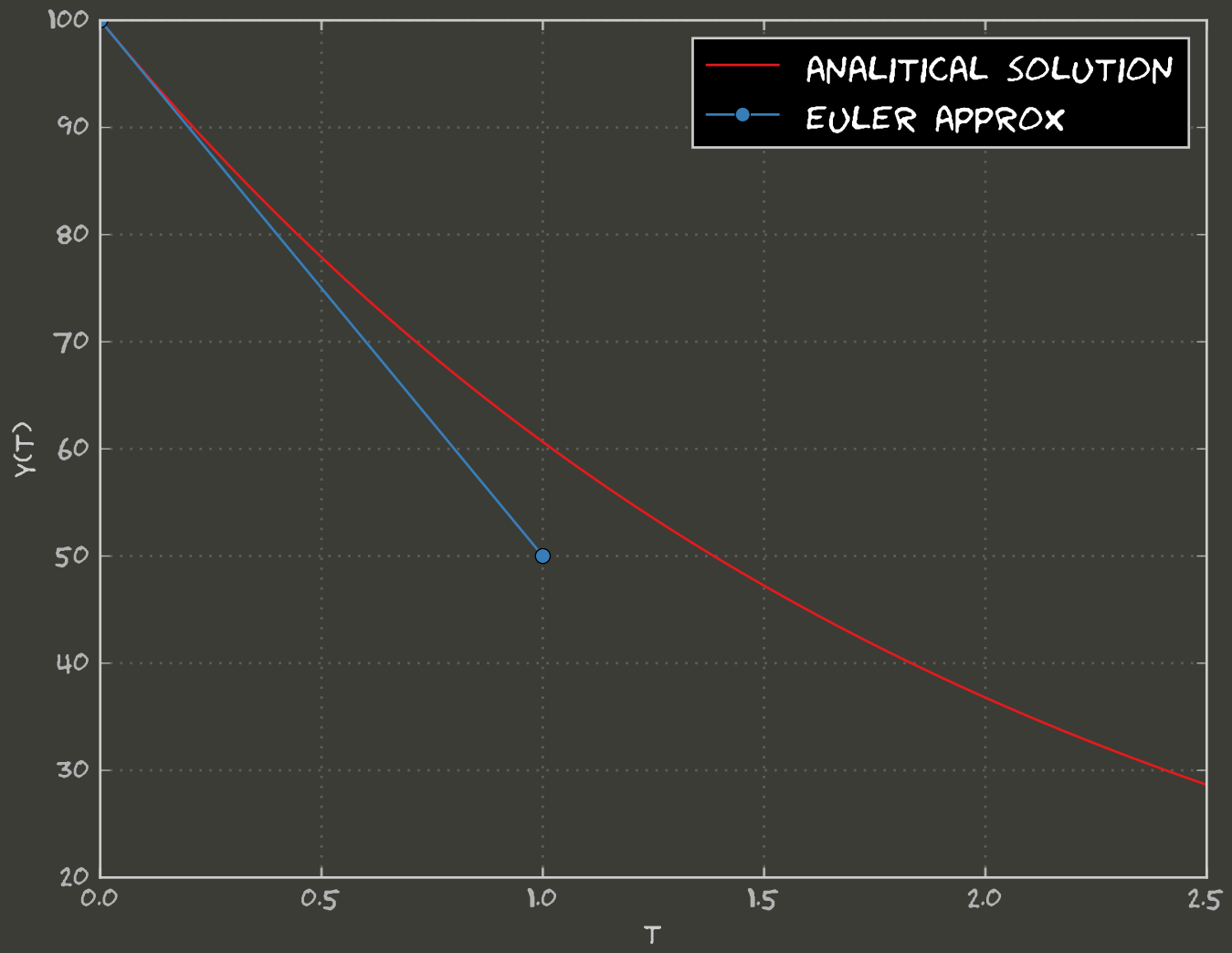
$k = 0.5$

$f(y,t) = -k\,y(t)$

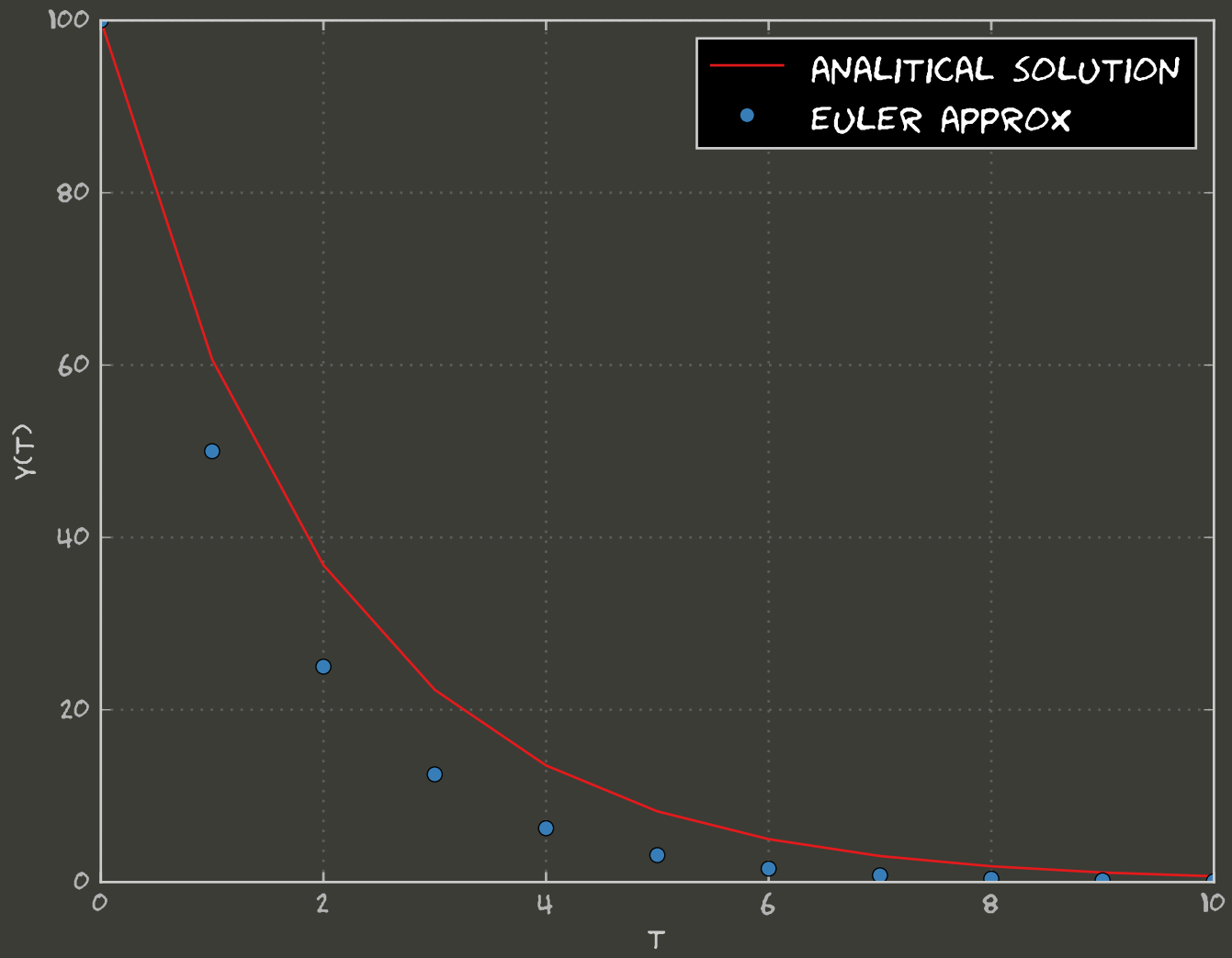
Euler's method: example

$$
\begin{aligned}
y_0 &= 100 \\
y_1 &= y_0 + f(t=0, y(t=0))\Delta t = \\
&= y_0 + (-k\, y_0)\, \Delta t = 100 - 0.5 \cdot 100 \cdot 0.1 = 95 \\
y_2 &= y_1 + f(t_1, y_1)\Delta t = 95 - 0.5 \cdot 95 \cdot 0.1 = 90.25 \\
y_3 &= y_2 + f(t_2, y_2)\Delta t = 90.25 - 0.5 \cdot 90.25 \cdot 0.1 = 85.74 \\
&\vdots
\end{aligned}
$$

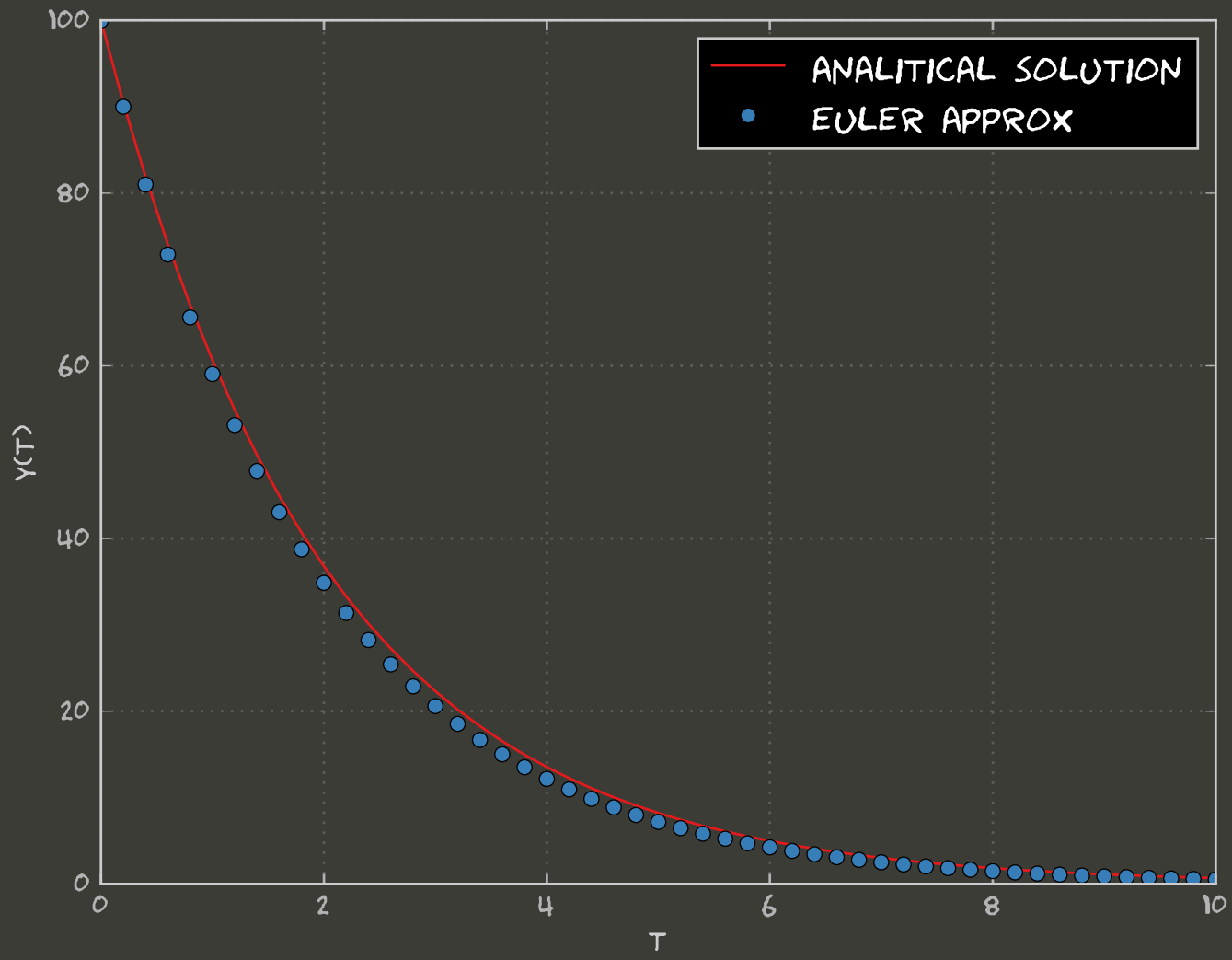
Euler's method: example



Euler's method: example



Euler's method: example



ODE: integrator

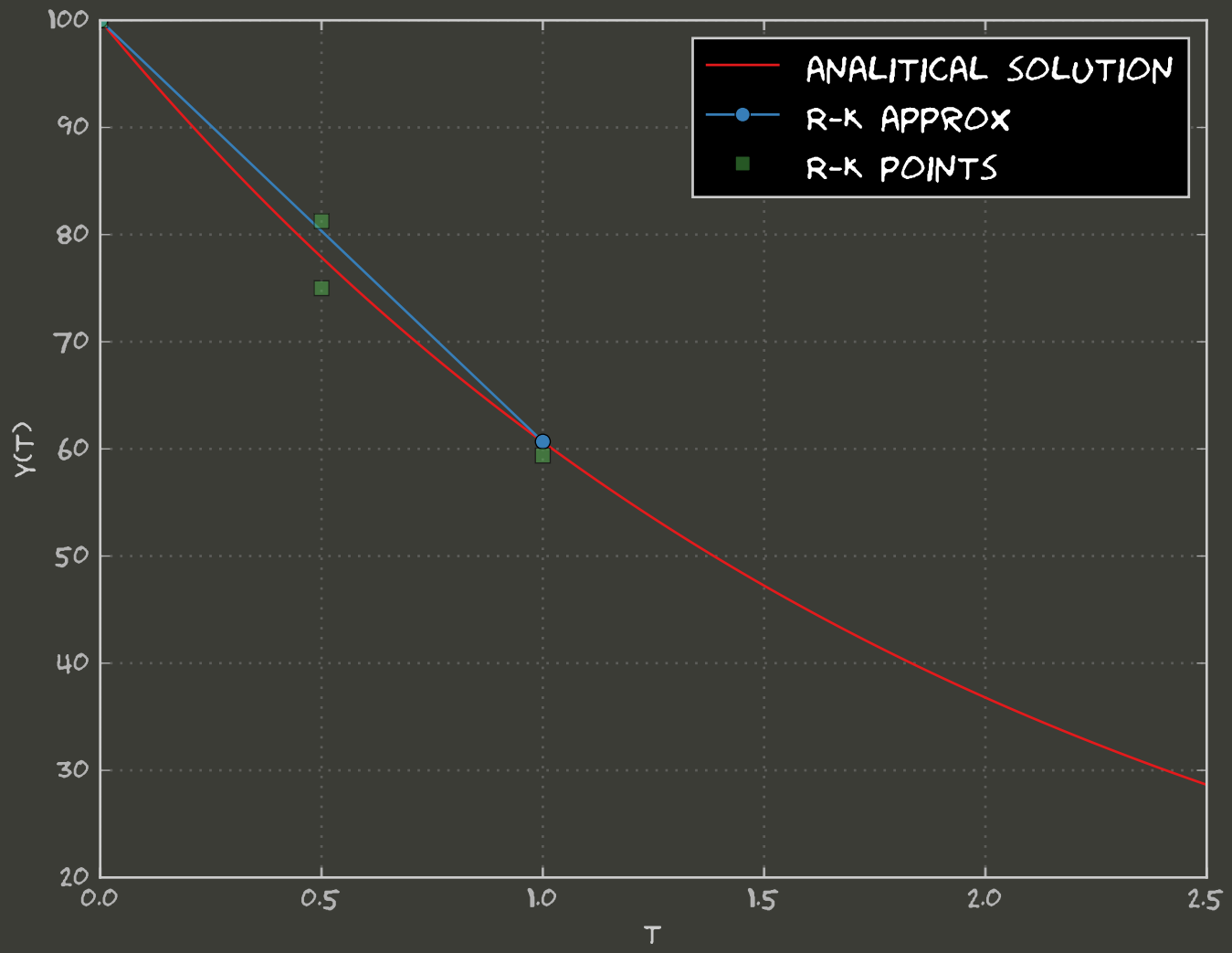
It is possible to increase the accuracy: 4^{th} order Runge Khutta method

- The independent variable is increased by a constant amount h:
- The dependent variables are increased by an amount that is a weighted average of 4 positions:

$$
\begin{aligned}
k_1 &= f(t_n, y_n) \\
k_2 &= f\left(t_n + \frac{h}{2}, y_n + k_1 \frac{h}{2}\right) \\
k_3 &= f\left(t_n + \frac{h}{2}, y_n + k_2 \frac{h}{2}\right) \\
k_4 &= f(t_n + h, y_n + k_3\, h)
\end{aligned}
$$

$$
y_{n+1} = y_n + h\left(\frac{k_1}{6} + \frac{k_2}{3} + \frac{k_3}{3} + \frac{k_4}{6}\right)
$$

Runge-Khutta: example



Runge-Khutta: example

Given:

$$\begin{cases} \frac{dy(t)}{dt} = -k\,y(t) \\ y(t=0) = 100 \end{cases}$$

with analytical solution $y(t) = 100\,e^{-k\,t}$

Parameters:

$\Delta t = 0.1$

$k = 0.5$

$f(y,t) = -k\,y(t)$

Runge-Khutta: example

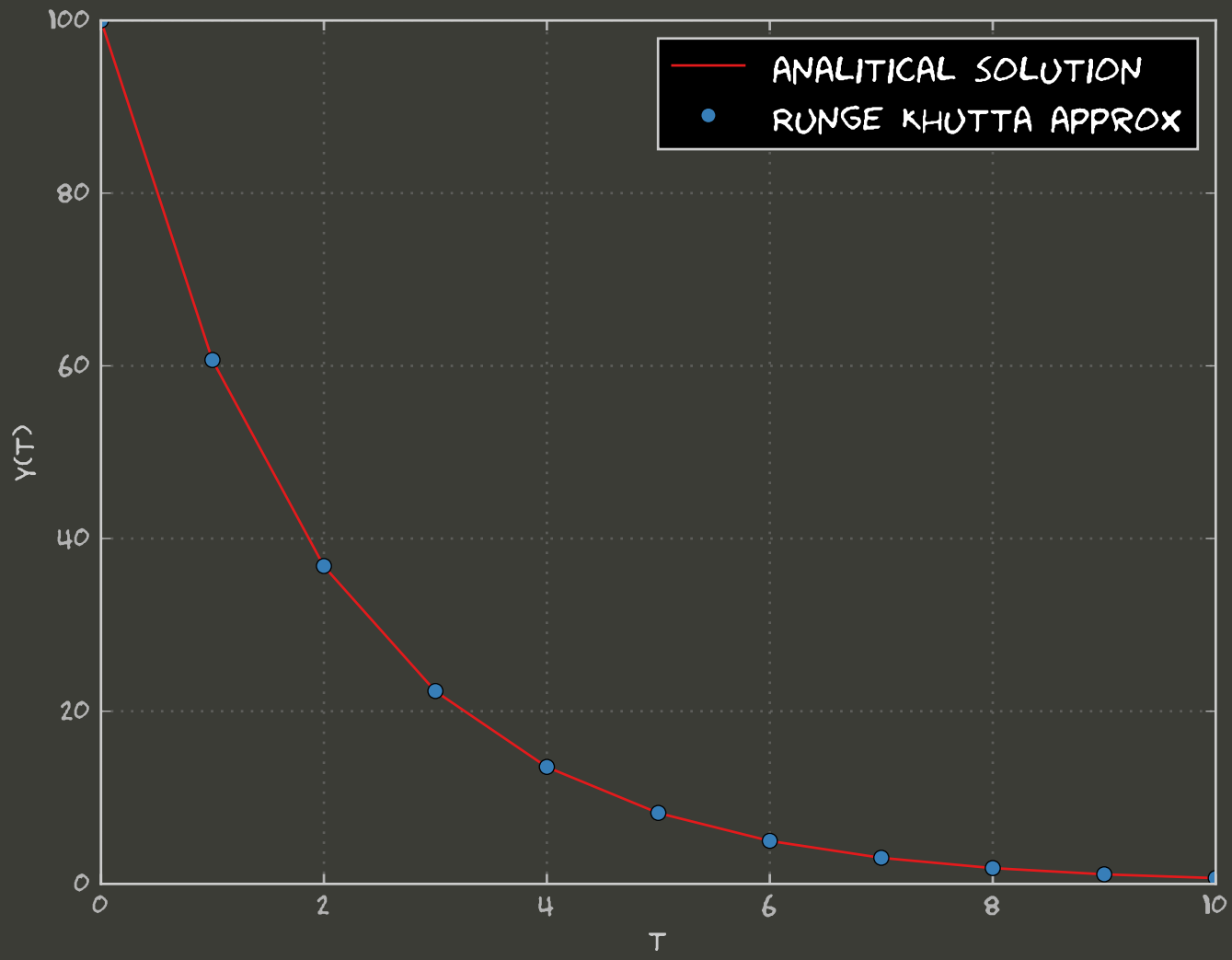
$$
\begin{aligned}
k_1 &= f(t_0,\, y_0) = -k\, y_0 = -0.5 \cdot 100 = -50 \\
k_2 &= f\left(t_0 + \frac{h}{2},\, y_n + k_1 \frac{h}{2}\right) = \\
&= f\left(100 - \frac{50}{2}\right) = -0.5 \cdot 75 = -37.5 \\
k_3 &= f\left(t_0 + \frac{h}{2},\, y_n + k_2 \frac{h}{2}\right) = \\
&= f\left(100 - \frac{37.5}{2}\right) = -0.5 \cdot 81.25 = -40.625 \\
k_4 &= f(t_0 + h,\, y_n + k_3\, h) \\
&= f(100 - 40.625) = -29.6875
\end{aligned}
$$

Runge-Khutta: example

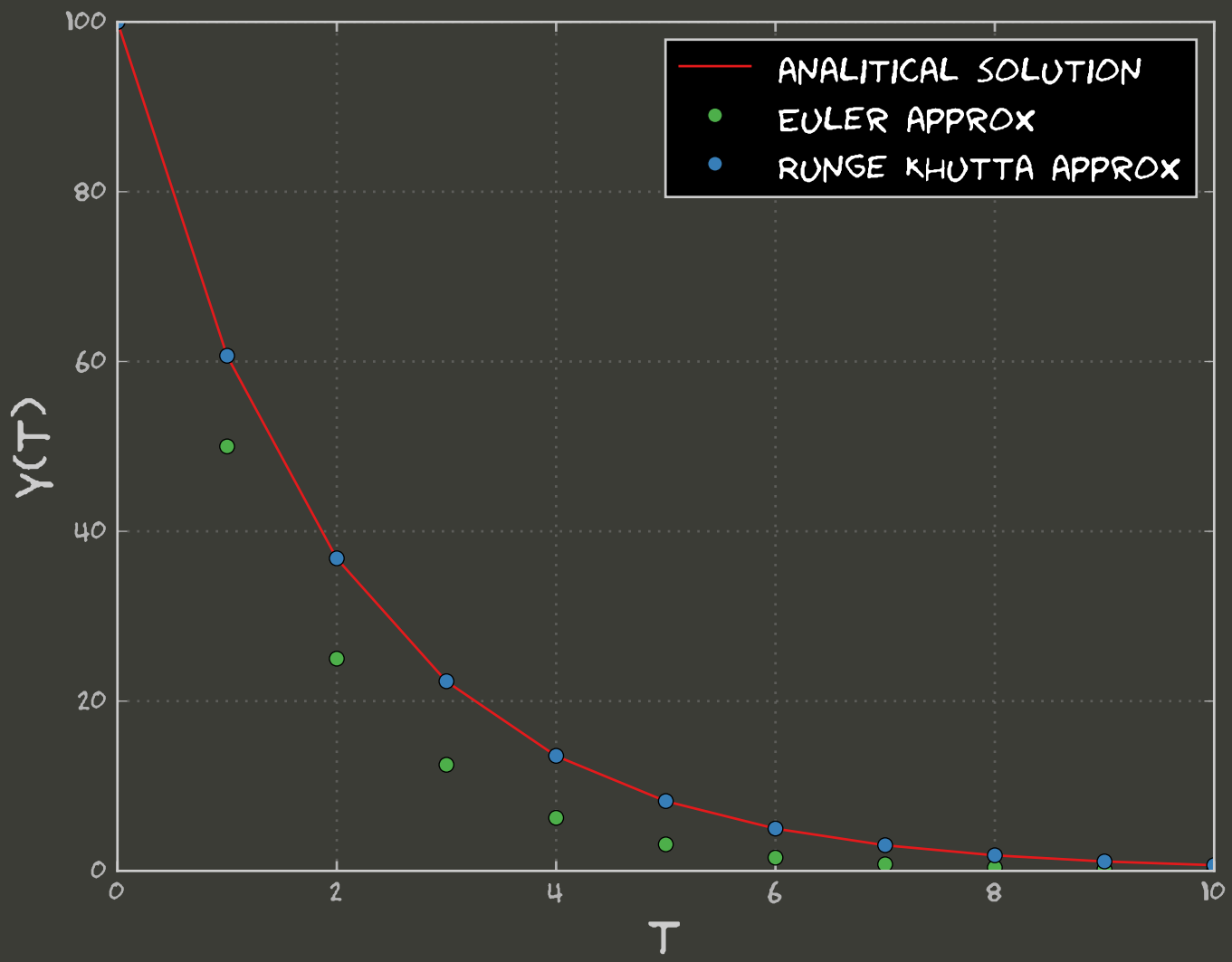
$$
\begin{aligned}
y_1 &= y_0 + \Delta t \left(\frac{k_1}{6} + \frac{k_2}{3} + \frac{k_3}{3} + \frac{k_4}{6} \right) = \\
&= 100 + 1 \cdot \left(-\frac{50}{6} - \frac{37.5}{3} - \frac{40.625}{3} - \frac{29.685}{6} \right) = \\
&= 60.6771
\end{aligned}
$$

while
$y(1) = 100 \cdot e^{-0.5 \cdot 1}$ = 60.6531

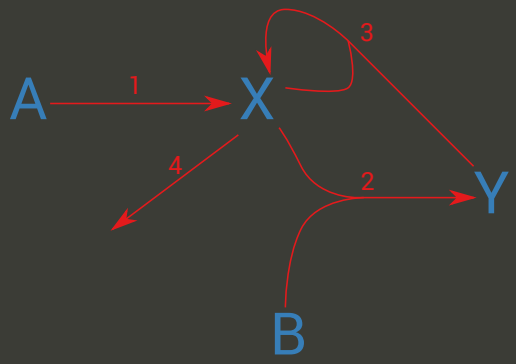
Runge-Khutta: example



Runge-Khutta: example



the BZ Reaction, the Brusselator & the Dynamics



$$A \longrightarrow X$$
$$B + X \longrightarrow Y + D$$
$$2X + Y \longrightarrow 3X$$
$$X \longrightarrow E$$

$$\begin{cases} \frac{\partial X}{\partial t} = k_1 A - k_2 B X + k_3 X^2 Y - k_4 X \\ \\ \frac{\partial Y}{\partial t} = k_2 B X - k_3 X^2 Y \end{cases}$$

The hidden hypothesis

$$\begin{cases} \frac{\partial X}{\partial t} = k_1 A - k_2 B X + k_3 X^2 Y - k_4 X \\ \\ \frac{\partial Y}{\partial t} = k_2 B X - k_3 X^2 Y \end{cases}$$

These differential equations have been derived according to the mass-action law

From rewriting rules to ODE: mass-action law

$$A \xrightarrow{k_1} X$$
$$B + X \xrightarrow{k_2} Y + D$$
$$2X + Y \xrightarrow{k_3} 3X$$
$$X \xrightarrow{k_4} E$$

$$\frac{dA}{dt} = -k_1[A]$$
$$\frac{dB}{dt} = -k_2[B][X]$$
$$\frac{dX}{dt} = k_1[A] - k_2[B][X] + k_3[X]^2[Y] - k_4[X]$$
$$\frac{dY}{dt} = k_2[B][X] - k_3[X]^2[Y]$$

From rewriting rules to ODE: how to automate

$$
\begin{aligned}
\alpha_{1,1}x_1 + \alpha_{1,2}x_2 + \cdots + \alpha_{1,N}x_N &\xrightarrow{k_1} \beta_{1,1}x_1 + \beta_{1,2}x_2 + \cdots + \beta_{1,N}x_N \\
\alpha_{2,1}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{2,N}x_N &\xrightarrow{k_2} \beta_{2,1}x_1 + \beta_{2,2}x_2 + \cdots + \beta_{2,N}x_N \\
\vdots \qquad\qquad & \qquad \vdots \\
\alpha_{M,1}x_1 + \alpha_{M,2}x_2 + \cdots + \alpha_{M,N}x_N &\xrightarrow{k_M} \beta_{M,1}x_1 + \beta_{M,2}x_2 + \cdots + \beta_{M,N}x_N
\end{aligned}
$$

$$
\begin{cases}
\frac{dx_1}{dt} = k_1 \delta_{1,1} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,1} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \cdots + \\
\qquad k_M \delta_{M,1} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}} \\
\frac{dx_2}{dt} = k_1 \delta_{1,2} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,2} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \cdots + \\
\qquad k_M \delta_{M,2} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}} \\
\qquad \vdots \\
\frac{dx_N}{dt} = k_1 \delta_{1,N} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,N} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \cdots + \\
\qquad k_M \delta_{M,N} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}}
\end{cases}
$$

From rewriting rules to ODE: how to automate

$$
\begin{cases}
\frac{dx_1}{dt} = k_1 \delta_{1,1} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,1} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \dots + \\
\qquad k_M \delta_{M,1} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}} \\
\frac{dx_2}{dt} = k_1 \delta_{1,2} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,2} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \dots + \\
\qquad k_M \delta_{M,2} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}} \\
\qquad \vdots \\
\frac{dx_N}{dt} = k_1 \delta_{1,N} [x_1]^{\alpha_{1,1}} [x_2]^{\alpha_{1,2}} \dots [x_N]^{\alpha_{1,N}} + \quad k_2 \delta_{2,N} [x_1]^{\alpha_{2,1}} [x_2]^{\alpha_{2,2}} \dots [x_N]^{\alpha_{2,N}} + \dots + \\
\qquad k_M \delta_{M,N} [x_1]^{\alpha_{M,1}} [x_2]^{\alpha_{M,2}} \dots [x_N]^{\alpha_{M,N}}
\end{cases}
$$

$$
\begin{cases}
\frac{dx_1}{dt} = \sum_i^M k_i \delta_{i,1} \prod_p^N [x_p]^{\alpha_{1,p}} \\
\frac{dx_2}{dt} = \sum_i^M k_i \delta_{i,2} \prod_p^N [x_p]^{\alpha_{2,p}} \\
\qquad \vdots \\
\frac{dx_N}{dt} = \sum_i^M k_i \delta_{i,N} \prod_p^N [x_p]^{\alpha_{N,p}}
\end{cases}
$$

From rewriting rules to ODE: how to automate

$$\frac{dx_j}{dt} = \sum_{i}^{M} k_i \, \delta_{i,j} \prod_{p}^{N} [x_p]^{\alpha_{i,p}}$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_M \end{pmatrix}$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} =$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

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$$X \xrightarrow{k_4} E$$

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0$$

From rewriting rules to ODE: Brusselator

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From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 - 1\, k_2\, x_1^0$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0$$

From rewriting rules to ODE: Brusselator

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 + k_3\, x_1^0$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

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$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2\, x_3^0$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

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$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1 - k_4\, x_1^0$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1 - k_4\, x_1^0\, x_2^1$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 \; - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 \; + \; k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1 - k_4\, x_1^0 \; x_2^1 x_3^0$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 + k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1 - k_4\, x_1^0\, x_2^1\, x_3^0\, x_4^0$$

From rewriting rules to ODE: Brusselator

$$A \xrightarrow{k_1} X$$

$$B + X \xrightarrow{k_2} Y + D$$

$$2X + Y \xrightarrow{k_3} 3X$$

$$X \xrightarrow{k_4} E$$

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = k_1\, x_1 - k_2\, x_2\, x_3 + k_3\, x_2^2\, x_4 - k_4\, x_2$$

From rewriting rules to ODE: draft algorithm

$$\frac{dx_j}{dt} = \sum_{i}^{M} k_i \, \delta_{i,j} \prod_{p}^{N} [x_p]^{\alpha_{i,p}}$$

It is possible to recognize three cycles:

- running over matrix D one column at a time
- running over matrix D and vector $\overrightarrow{k}$ one row at a time
- running over matrix L one column at a time

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[X]}{dt} = 1\, k_1 x_1^1\, x_2^0\, x_3^0\, x_4^0 - 1\, k_2\, x_1^0\, x_2^1\, x_3^1\, x_4^0 + k_3\, x_1^0\, x_2^2\, x_3^0\, x_4^1 - k_4\, x_1^0\; x_2^1\, x_3^0\, x_4^0$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^1\, x_2{}^0\, x_3{}^0\, x_4{}^0$$

$$-1\, k_2 \quad x_1{}^0\, x_2{}^1\, x_3{}^1\, x_4{}^0$$

$$+1\, k_3 \quad x_1{}^0\, x_2{}^2\, x_3{}^0\, x_4{}^1$$

$$-1\, k_4 \quad x_1{}^0\, x_2{}^1\, x_3{}^0\, x_4{}^0$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^{1}\, x_2{}^{0}\, x_3{}^{0}\, x_4{}^{0}$$

$$-1\, k_2 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{1}\, x_4{}^{0}$$

$$+1\, k_3 \quad x_1{}^{0}\, x_2{}^{2}\, x_3{}^{0}\, x_4{}^{1}$$

$$-1\, k_4 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{0}\, x_4{}^{0}$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^1\, x_2{}^0\, x_3{}^0\, x_4{}^0$$

$$-1\, k_2 \quad x_1{}^0\, x_2{}^1\, x_3{}^1\, x_4{}^0$$

$$+1\, k_3 \quad x_1{}^0\, x_2{}^2\, x_3{}^0\, x_4{}^1$$

$$-1\, k_4 \quad x_1{}^0\, x_2{}^1\, x_3{}^0\, x_4{}^0$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^{1}\, x_2{}^{0}\, x_3{}^{0}\, x_4{}^{0}$$

$$-1\, k_2 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{1}\, x_4{}^{0}$$

$$+1\, k_3 \quad x_1{}^{0}\, x_2{}^{2}\, x_3{}^{0}\, x_4{}^{1}$$

$$-1\, k_4 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{0}\, x_4{}^{0}$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^1\, x_2{}^0\, x_3{}^0\, x_4{}^0$$

$$-1\, k_2 \quad x_1{}^0\, x_2{}^1\, x_3{}^1\, x_4{}^0$$

$$+1\, k_3 \quad x_1{}^0\, x_2{}^2\, x_3{}^0\, x_4{}^1$$

$$-1\, k_4 \quad x_1{}^0\, x_2{}^1\, x_3{}^0\, x_4{}^0$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^1\, x_2{}^0\, x_3{}^0\, x_4{}^0$$

$$-1\, k_2 \quad x_1{}^0\, x_2{}^1\, x_3{}^1\, x_4{}^0$$

$$+1\, k_3 \quad x_1{}^0\, x_2{}^2\, x_3{}^0\, x_4{}^1$$

$$-1\, k_4 \quad x_1{}^0\, x_2{}^1\, x_3{}^0\, x_4{}^0$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^{1}\, x_2{}^{0}\, x_3{}^{0}\, x_4{}^{0}$$

$$-1\, k_2 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{1}\, x_4{}^{0}$$

$$+1\, k_3 \quad x_1{}^{0}\, x_2{}^{2}\, x_3{}^{0}\, x_4{}^{1}$$

$$-1\, k_4 \quad x_1{}^{0}\, x_2{}^{1}\, x_3{}^{0}\, x_4{}^{0}$$

```
np.power(x, leftSide)
```

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^1 \cdot x_2{}^0 \cdot x_3{}^0 \cdot x_4{}^0$$

$$-1\, k_2 \quad x_1{}^0 \cdot x_2{}^1 \cdot x_3{}^1 \cdot x_4{}^0$$

$$+1\, k_3 \quad x_1{}^0 \cdot x_2{}^2 \cdot x_3{}^0 \cdot x_4{}^1$$

$$-1\, k_4 \quad x_1{}^0 \cdot x_2{}^1 \cdot x_3{}^0 \cdot x_4{}^0$$

```
np.power(x, leftSide)
```

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$+1\, k_1 \quad x_1{}^{1} \cdot x_2{}^{0} \cdot x_3{}^{0} \cdot x_4{}^{0}$$

$$-1\, k_2 \quad x_1{}^{0} \cdot x_2{}^{1} \cdot x_3{}^{1} \cdot x_4{}^{0}$$

$$+1\, k_3 \quad x_1{}^{0} \cdot x_2{}^{2} \cdot x_3{}^{0} \cdot x_4{}^{1}$$

$$-1\, k_4 \quad x_1{}^{0} \cdot x_2{}^{1} \cdot x_3{}^{0} \cdot x_4{}^{0}$$

```
np.product(np.power(x, leftSide),
axis=1)
```

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$+1\, k_1 \quad \tau_1$

$-1\, k_2 \quad \tau_2$

```
np.product(np.power(x, leftSide), axis=1)
```

$+1\, k_3 \quad \tau_3$

$-1\, k_4 \quad \tau_4$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$+1\, k_1 \quad \tau_{2,1}$

$-1\, k_2 \quad \tau_{2,2}$

$+1\, k_3 \quad \tau_{2,3}$

$-1\, k_4 \quad \tau_{2,4}$

```
np.product(np.power(x, leftSide),
axis=1)
```

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$+1\, k_1 \quad \tau_{2,1}$

$-1\, k_2 \quad \tau_{2,2}$

$+1\, k_3 \quad \tau_{2,3}$

$-1\, k_4 \quad \tau_{2,4}$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$+1\, k_1 \quad \tau_{2,1}$

$-1\, k_2 \quad \tau_{2,2}$

$+1\, k_3 \quad \tau_{2,3}$

$-1\, k_4 \quad \tau_{2,4}$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[A]}{dt} = -1k_1[A]$$

$$\frac{d[X]}{dt} = +1k_1[A] - 1k_2[B][X] + 1k_3[X]^2[Y] - 1k_4[X]$$

$$\frac{d[B]}{dt} = -1k_2[B][X]$$

$$\frac{d[Y]}{dt} = +1k_2[B][X] - 1k_3[X]^2[Y]$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$D^T * \overrightarrow{k}$ component-wise

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$D^T * \overrightarrow{k}$ component-wise

```
D.T * K
```

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \overrightarrow{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

$$\frac{d[A]}{dt} = -1k_1[A]$$

$$\frac{d[X]}{dt} = +1k_1[A] - 1k_2[B][X] + 1k_3[X]^2[Y] - 1k_4[X]$$

$$\frac{d[B]}{dt} = -1k_2[B][X]$$

$$\frac{d[Y]}{dt} = +1k_2[B][X] - 1k_3[X]^2[Y]$$

Automatic generation of ODE: exploiting numPy

$$D = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 \end{pmatrix} L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \vec{k} = \begin{pmatrix} k_1 \\ k_2 \\ k_3 \\ k_4 \end{pmatrix}$$

```
(D.T * K).dot(tau)
```

Automatic generation of ODE: exploiting numPy

```
def myCauchy(x, t):
    terms = np.product(np.power(x, leftSide), axis=1)
    teta = delta.T * aConstants
    dx = teta.dot(terms)
    for spec in dictFeedSpecs:
        dx[spec] = 0.0
    return dx
```

the Brusselator Dynamics: ODE